CLINICAL · EAR

Vertigo: BPPV, neuritis and central causes

Vertigo is the illusion of movement, usually spinning, and in most patients it comes from the inner ear. Peripheral causes account for about 85%, but the duty of the doctor is to catch the dangerous central minority. Four questions sort most cases: how long the attacks last, what triggers them, whether hearing is affected, and whether there are neurological signs.

AT A GLANCE

- Timing, triggers, hearing and neurology sort most cases.
- BPPV is seconds; Ménière's is hours; neuritis is days.
- HINTS and focal signs catch stroke.
- Epley for BPPV; early rehabilitation for neuritis.
- Horizontal canal BPPV: supine roll test and barbecue roll or Gufoni; anterior canal: Yacovino manoeuvre.

1 CHAPTER ONE Overview

A symptom, not a diagnosis.

1.1 Definitions

1.1.1 Vertigo

A false sense of rotation or movement of oneself or the surroundings, from asymmetry in the vestibular system.

1.1.2 Other kinds of dizziness

- **Disequilibrium:** unsteadiness without a sense of rotation.
- **Oscillopsia:** the visual world jumps with head movement, typical of bilateral vestibular loss.
- **Presyncope:** light-headedness from reduced cerebral perfusion (cardiac, orthostatic, hypoglycaemic).
- **Physiological:** motion sickness and height vertigo.

1.2 Peripheral or central

CAUSES OF VERTIGO

- PERIPHERAL **BPPV, vestibular neuritis** labyrinthitis, Ménière's disease, perilymph fistula, superior canal dehiscence, ototoxicity, trauma, syphilis
- CENTRAL **Posterior circulation stroke** vertebrobasilar insufficiency, Wallenberg syndrome, vestibular migraine, multiple sclerosis, cerebellar and brainstem tumours, epilepsy
- OTHER **Cervical** ocular, psychogenic, cardiovascular

2 CHAPTER TWO The history

Timing is everything.

2.1 Duration

DURATION OF ATTACKS

DURATION	TYPICAL CAUSES
Seconds	Benign paroxysmal positional vertigo
Minutes	Vertebrobasilar ischaemia, vestibular migraine
Minutes to hours	Ménière's disease, recurrent vestibulopathy, vestibular migraine
Days to weeks	Vestibular neuritis, labyrinthitis, stroke
Variable	Perilymph fistula, superior canal dehiscence

2.2 Triggers

Head position (BPPV), loud sound or pressure (superior canal dehiscence, fistula), straining, neck movement, standing up (orthostatic hypotension), and migraine triggers.

2.3 Hearing

Hearing loss or tinnitus points to the labyrinth (labyrinthitis, Ménière's disease, fistula) or the cerebellopontine angle, or to an AICA stroke.

2.4 Neurological symptoms

Double vision, slurred speech, numbness, weakness, difficulty swallowing, new severe headache: central until proven otherwise.

3 CHAPTER THREE Examination

Eyes, head and gait.

3.1 Nystagmus

Look in the primary position and on gaze, with and without fixation (Frenzel glasses). Peripheral nystagmus is horizontal-torsional, beats away from the affected side and is suppressed by fixation; it is stronger looking towards the fast phase (*Alexander's law*).

3.2 Head impulse test

A rapid head turn: a catch-up saccade means the vestibulo-ocular reflex on that side has failed, a peripheral lesion.

3.3 HINTS

In acute continuous vertigo with nystagmus: a *normal* head impulse, direction-changing nystagmus, or a vertical skew deviation points to a stroke.

3.4 Positional tests

The Dix-Hallpike manoeuvre for posterior canal BPPV, and the supine roll test for the

lateral canal.

3.5 Other bedside tests

Romberg and tandem gait, head shake, dynamic visual acuity (losing more than five lines suggests bilateral loss), fistula test, and a full neurological and cranial nerve examination.

PERIPHERAL OR CENTRAL FINDINGS

FINDING	ACUTE PERIPHERAL	CENTRAL
Nystagmus	Mixed horizontal-torsional	Pure vertical or torsional, direction-changing
Fixation	Suppresses	Does not suppress
Head impulse	Abnormal	Normal
Pursuit	Normal	Saccadic
Saccades	Normal	Dysmetric
Gait	Can walk	Often cannot stand
Other CNS signs	Absent	Often present

4 CHAPTER FOUR Peripheral causes

The common ones.

4.1 Benign paroxysmal positional vertigo

4.1.1 Mechanism

Otoconia dislodged from the utricle settle in a semicircular canal, most often the posterior canal, and move with gravity when the head changes position (*canalolithiasis*). It often follows head injury or ear infection, or occurs in older people without cause.

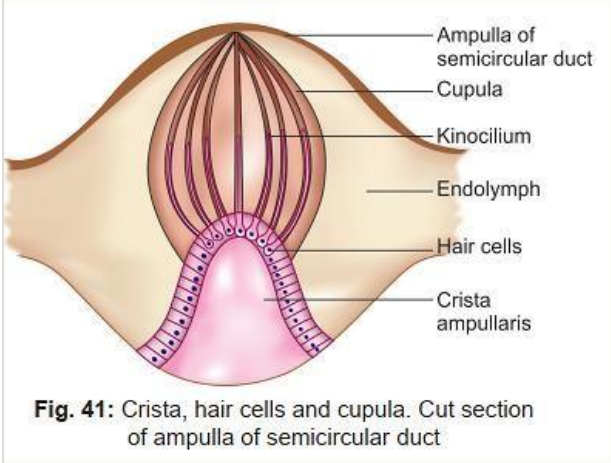


Fig. 41: Crista, hair cells and cupula. Cut section of ampulla of semicircular duct

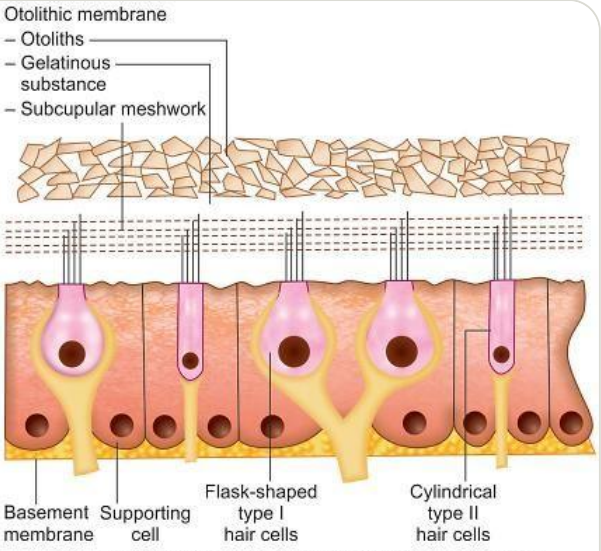


Fig. 43: Macula of otolith organs: utricle and saccule

FIGURE 1 The crista ampullaris with its cupula, and the macula of the utricle and saccule whose otoconia cause BPPV.

Anatomy atlas · ENT Steps p. 78, ENT Steps p. 80

4.1.2 Picture

Brief vertigo, under a minute, on lying down, rolling over or looking up. No hearing loss.

4.1.3 Dix-Hallpike nystagmus

Rotatory and geotropic, after a latency of a few seconds, rising then fading within a minute, reversing on sitting up, and fatiguing on repetition.

4.1.4 Treatment

The *Epley manoeuvre* repositions the debris into the utricle; about 80% are cured by one manoeuvre. Semont manoeuvre and Brandt-Daroff exercises are alternatives. Most cases resolve in months anyway. Surgery (posterior canal occlusion or singular neurectomy) is rarely needed.

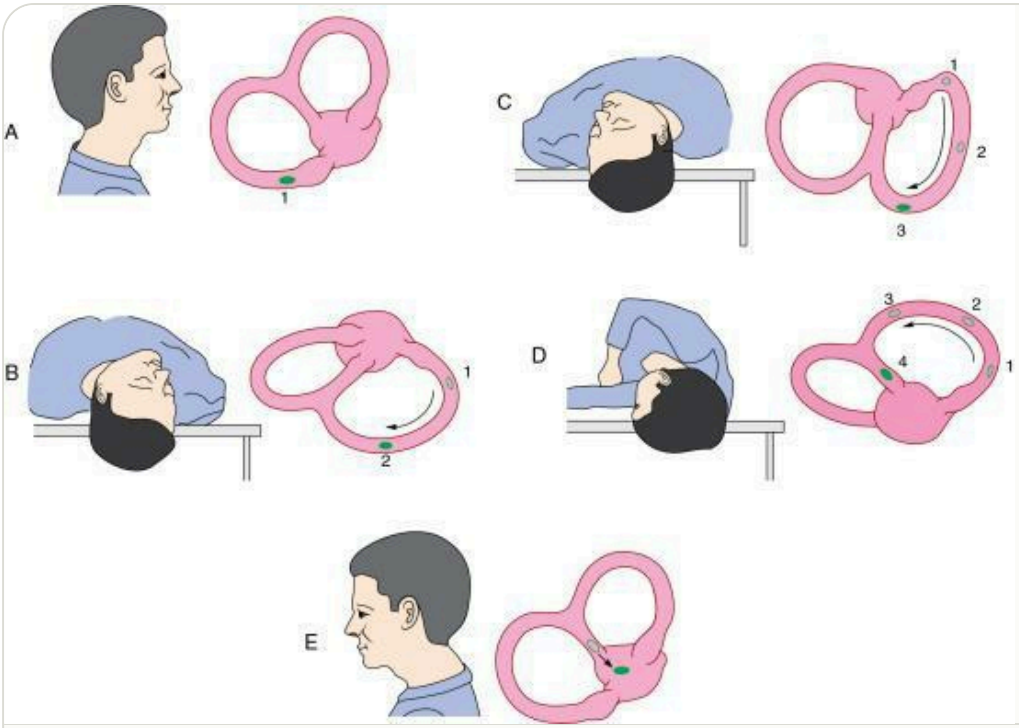


FIGURE 2 The Epley manoeuvre for posterior canal BPPV, with the position of the patient and of the otolith debris in the canal at each step.

Dhingra, fig. 7.1

BEYOND THE POSTERIOR CANAL

The Oxford Textbook gives the current picture. The **posterior canal** causes about 90 to 95% of BPPV, because it is the lowest canal when lying down; the **horizontal (lateral) canal** causes 5 to 15%, and the anterior canal very few. Repositioning cures about 80% after one session and over 90% after up to four, and patients can be taught to do it themselves. No postural restrictions are needed afterwards, because they do not prevent recurrence. Recurrence is common, in up to a third of patients within four years, and the occasional Epley manoeuvre converts posterior into horizontal canal BPPV.

Horizontal canal BPPV is diagnosed with the supine roll test, which produces horizontal nystagmus on turning to either side. In the commoner *geotropic* form (beating towards the ground, from free particles in the canal), the side with the stronger nystagmus is the affected ear. In the *apogeotropic* form (beating away from the ground, thought to be cupulolithiasis or particles near the cupula), the weaker side is the affected ear. It is treated with the **Lempert 360° (barbecue or log) roll** towards the healthy side, the **Gufoni manoeuvre**, or prolonged lying on one side; these work moderately well for the geotropic form, and the evidence for the apogeotropic form is weak. **Anterior canal BPPV** produces down-beating nystagmus with a torsional component and can be treated by the **Yacovino (deep head-hanging) manoeuvre**, which treats either side because the head is not

turned; down-beating positional nystagmus that does not settle must be imaged to exclude a cerebellar lesion.

Vestibular sedatives and benzodiazepines slow central compensation and increase falls in the elderly, who carry a much higher fall risk when dizzy, so they have no role in BPPV. Atypical nystagmus, failure after several manoeuvres and frequent recurrence call for further investigation.

POSITIONAL VERTIGO: PERIPHERAL OR CENTRAL

FEATURE	PERIPHERAL	CENTRAL
Latency	Present	Often none
Duration	Under a minute	Persists
Fatigue	Yes	No
Direction	Fixed, torsional	Changing or vertical
Vertigo	Marked	Often mild

4.2 Vestibular neuritis

4.2.1 Mechanism

Probably viral inflammation of the vestibular nerve or ganglion.

4.2.2 Picture

Sudden, severe, continuous vertigo with nausea and vomiting, lasting days and settling over two to three weeks, without hearing loss. Nystagmus beats away from the affected side; the head impulse is abnormal towards it; calorics are reduced. BPPV follows in up to 15%.

4.2.3 Treatment

Short-course vestibular sedatives, possibly steroids early, then early vestibular rehabilitation. Failure to improve at two to three weeks means a central lesion, such as a cerebellar infarct, must be excluded.

4.3 Labyrinthitis

As neuritis, but with hearing loss, from viral or bacterial infection of the labyrinth itself.

4.4 Ménière's disease

Attacks of vertigo lasting hours with fluctuating hearing loss, tinnitus and fullness.

4.5 Recurrent vestibulopathy

Ménière-like spells without hearing symptoms. About 60% remit; a minority become Ménière's disease or BPPV.

4.6 Perilymph fistula

Episodic vertigo and fluctuating hearing after stapes surgery, barotrauma, heavy lifting or coughing. Bed rest with the head up and laxatives; explore if worsening.

4.7 Superior canal dehiscence

Vertigo provoked by sound and pressure, autophony and a low-frequency conductive loss with intact reflexes (see the separate topic).

4.8 Bilateral vestibular loss

Oscillopsia and unsteadiness in the dark, without true vertigo, most often from aminoglycosides. Cerebellar ataxia with bilateral vestibular loss is called *CABV*, or *CANVAS* when sensory neuropathy is added.

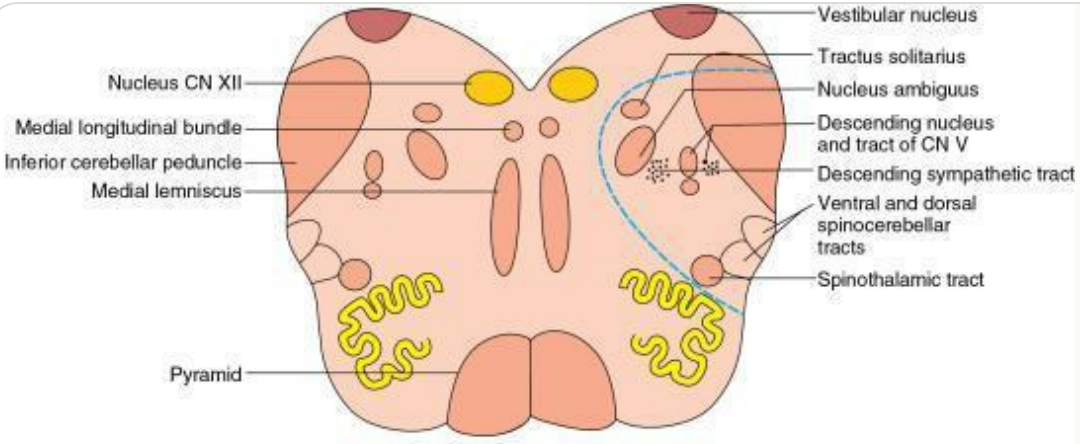
5 CHAPTER FIVE Central causes

The ones not to miss.

5.1 Posterior circulation stroke

POSTERIOR CIRCULATION STROKE SYNDROMES

FEATURE	PICA	AICA	SCA
Vertigo, nystagmus, ataxia	Yes	Yes	Yes
Hearing loss, tinnitus	No	Yes	No
Facial palsy	Sometimes	Yes	No
Dysphagia, hoarseness	Yes	No	No
Contralateral loss of vibration and position	No	No	Yes



• Inferior cerebellar peduncle	Vertigo, nausea, vomiting and nystagmus
• Spinocerebellar tracts	Ataxia
• Nucleus ambiguus (CN X, IX)	Hoarseness and dysphagia
• Descending sympathetic tract	Horner syndrome
• Uncrossed fibres of spinothalamic tract	Loss of pain and temperature on ipsilateral face
• Descending nucleus and tract of CN V	Pain and numbness over ipsilateral face
• Crossed fibres of spinothalamic tract	Contralateral loss of pain and temperature of arm, trunk and leg

FIGURE 3 Lateral medullary (Wallenberg) syndrome from occlusion of the posterior inferior cerebellar artery.

Dhingra, fig. 7.2

5.1.1 Wallenberg syndrome

Violent vertigo with hoarseness, dysphagia, ipsilateral Horner syndrome, ipsilateral facial and contralateral body loss of pain and temperature, and ataxia.

5.2 Vertebrobasilar insufficiency

Attacks lasting minutes in patients over 50, with visual disturbance, diplopia or drop attacks, sometimes on neck movement.

5.3 Vestibular migraine

Attacks of vertigo, minutes to hours, with or near migraine headache, often in young women with a family history; response to migraine treatment supports the diagnosis.

5.4 Multiple sclerosis

Young adults with vertigo and pendular or dissociated nystagmus, and other neurological episodes.

5.5 Other central causes

Cerebellar and brainstem tumours, Chiari malformation, idiopathic intracranial hypertension, encephalitis and epilepsy.

6 CHAPTER SIX Other causes

When it isn't the ear or the brain.

6.1 Cervical vertigo

Vertigo with neck movement after a neck injury; controversial and probably over-diagnosed.

6.2 Cardiovascular

Presyncope from arrhythmia, orthostatic hypotension or hypoglycaemia; not true

rotatory vertigo.

6.3 Psychogenic

Vague floating or swaying, with anxiety, no nystagmus and normal tests. Persistent postural-perceptual dizziness often follows an acute vestibular event.

7 CHAPTER SEVEN Management

Principles.

ACUTE VERTIGO ON CALL

1. Ask about timing, triggers, hearing and neurological symptoms.
2. Examine nystagmus, head impulse, skew and gait (HINTS).
3. Any focal sign, new severe headache, inability to stand, or a normal head impulse with nystagmus: treat as stroke until proven otherwise.
4. Dix-Hallpike only once the neck and the patient's condition allow it.
5. Give antiemetics or vestibular sedatives for a few days only.

7.1 Vestibular rehabilitation

Head and eye exercises that drive central compensation, the mainstay after any peripheral loss.

7.2 Avoid long-term sedatives

They slow compensation.

PEARLS

- An acoustic neuroma usually causes unsteadiness, not severe vertigo.
- Fatigable, delayed, torsional positional nystagmus is peripheral; immediate, persistent or vertical nystagmus is central.

- A positive fistula test: perilymph fistula, cholesteatoma with a fistula, Ménière's disease, superior canal dehiscence, or syphilis.

OPERATIVE GUIDES IN THEATRE 2 operations

Surgery for Vertigo
Surgery for vertigo
Superior Semicircular Canal Dehiscence
Superior canal dehiscence

8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Peripheral versus central** vertigo and nystagmus.
- **BPPV:** mechanism, Dix-Hallpike findings, Epley manoeuvre.
- **Vestibular neuritis** versus labyrinthitis.
- **HINTS** in acute vertigo.
- **Wallenberg syndrome.**

8.2 Common pitfalls

COMMON PITFALLS

- Labelling a stroke as neuritis because the patient is vomiting.
- Missing a normal head impulse in continuous vertigo.
- Keeping patients on vestibular sedatives for weeks.
- Diagnosing BPPV without seeing the nystagmus.

Written for this app from Oxford Textbook of Otolaryngology ch. 7, 15, Oxford Specialist Handbook ch. 18, Oxford Handbook of ENT ch. 7, Dhingra ch. 7 and K. J. Lee ch. 16. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Dural Sinus Thrombosis

IMAGING FINDINGS

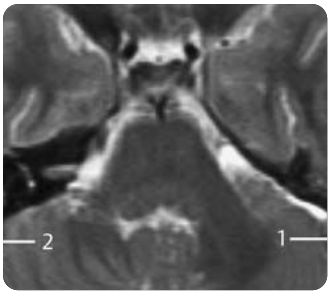
- Filling defect in the sigmoid or transverse sinus on contrast CT or MR venography
- Empty delta sign; dense sinus on non-contrast CT
- Adjacent mastoiditis with bone erosion over the sinus

TOP DIFFERENTIALS

- Arachnoid granulation (round, CSF signal)
- Hypoplastic transverse sinus
- Slow flow artefact

CHECKLIST

- Extension to the jugular vein and other sinuses
- Venous infarction and raised intracranial pressure



MRI, T2-weighted, axial. Otomastoiditis can spread intracranially, resulting in dural sinus thrombophlebitis, which may compromise the sigmoid and…



Axial CECT in a patient with postauricular tender mass, headache, and fever reveals postauricular abscess and coalescent…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1042.